

DABAA, Egypt

WMO: 623090

Lat: 31.0327N

Lon: 28.4448E

Elev: 59

StdP: 14.66

Time Zone: 2.00 (E02)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	45.9	47.5	27.2	21.2	62.3	31.6	26.0	60.1	40.5	51.8	30.4	55.0	7.4	240	0.351

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.9	92.3	69.9	89.2	73.2	87.7	74.1	77.9	85.6	77.0	84.9	76.2	84.2	8.8	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.6	134.3	82.2	74.6	129.6	82.0	73.6	125.3	81.7	41.4	85.6	40.5	84.8	39.7	84.4	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.8	20.3	16.7	DB	42.7	103.9	1.6	5.3	41.5	107.7	40.6	110.8	39.7	113.7	38.5	117.6	(4)
(5)			WB	37.9	79.8	2.6	1.7	36.0	81.1	34.5	82.1	33.0	83.1	31.1	84.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	69.7	57.2	58.8	62.0	65.9	71.1	76.4	80.3	81.4	79.3	74.4	67.8	60.8	(6)
(7)		DBStd	9.13	3.46	4.43	4.75	4.46	4.47	3.23	2.36	1.93	2.55	3.25	3.78	3.49	(7)
(8)		HDD50	1	1	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	733	242	183	120	35	1	0	0	0	0	0	14	138	(9)
(10)		CDD50	7183	225	246	371	477	654	792	939	975	880	755	535	334	(10)
(11)		CDD65	2440	1	8	26	62	190	342	474	510	430	291	100	6	(11)
(12)		CDH74	21089	16	76	229	386	1054	2643	4879	5715	4093	1708	272	18	(12)
(13)		CDH80	6418	2	23	88	139	324	664	1598	2048	1207	298	24	3	(13)
(14)	Wind	WSAvg	6.9	7.9	8.2	7.9	7.6	7.1	7.0	7.5	6.3	6.0	5.6	5.4	6.8	(14)
(15)	Precipitation	PrecAvg	6.8	1.7	1.2	0.6	0.2	0.1	0.0	0.0	0.0	0.0	0.3	0.9	1.7	(15)
(16)		PrecMax	10.7	3.3	2.7	1.7	0.8	0.5	0.1	0.0	0.0	0.2	1.2	3.4	3.6	(16)
(17)		PrecMin	3.5	0.7	0.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.4	(17)
(18)		PrecStd	1.6	0.7	0.6	0.4	0.2	0.1	0.0	0.0	0.0	0.1	0.3	0.7	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.9	83.3	90.0	91.6	96.8	95.2	95.3	93.5	94.9	90.8	82.7	75.9	(19)
(20)			MCWB	56.6	58.8	60.8	62.1	65.9	68.7	72.6	74.4	70.9	68.1	65.1	60.0	(20)
(21)		2%	DB	69.6	74.5	80.1	83.8	87.7	87.9	89.5	89.6	89.2	85.5	79.1	71.5	(21)
(22)			MCWB	55.5	55.6	58.5	61.0	65.1	71.1	75.0	76.1	73.6	69.6	66.8	60.0	(22)
(23)		5%	DB	66.6	69.3	73.6	77.8	82.3	85.4	87.8	88.3	87.1	82.8	77.0	69.6	(23)
(24)			MCWB	55.7	55.1	58.0	61.4	66.2	72.0	74.9	75.8	73.9	70.3	65.3	59.1	(24)
(25)		10%	DB	64.8	66.2	69.4	73.5	78.9	83.6	86.4	87.3	85.5	81.0	75.0	68.0	(25)
(26)			MCWB	55.2	55.6	58.2	61.7	66.5	71.7	74.5	75.4	73.4	69.7	64.6	58.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.9	62.4	64.3	66.5	71.1	76.1	78.6	79.5	77.8	75.0	71.9	65.6	(27)
(28)			MCDWB	67.7	69.8	75.6	76.5	79.8	83.9	85.8	87.1	86.0	82.1	77.7	72.0	(28)
(29)		2%	WB	58.8	60.1	62.2	65.2	69.8	74.9	77.3	78.2	76.5	73.2	69.4	62.3	(29)
(30)			MCDWB	65.6	67.4	70.5	74.4	78.3	82.3	85.0	85.8	84.5	81.1	76.2	69.5	(30)
(31)		5%	WB	57.4	58.4	60.8	63.9	68.8	74.1	76.4	77.4	75.4	71.8	67.6	60.5	(31)
(32)			MCDWB	64.7	65.7	69.1	72.6	77.1	81.4	84.3	85.2	83.8	80.0	74.9	67.5	(32)
(33)		10%	WB	56.0	57.0	59.5	62.8	67.8	73.1	75.6	76.5	74.4	70.7	65.6	59.1	(33)
(34)			MCDWB	63.4	64.5	67.4	71.0	75.9	80.5	83.5	84.6	83.0	79.1	73.1	66.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	14.8	15.0	15.1	15.1	15.3	13.8	12.8	12.9	13.1	13.4	13.7	14.2	(35)
(36)			MCDBR	18.2	21.2	23.5	23.9	23.5	17.2	15.2	14.5	16.0	16.5	16.6	16.6	(36)
(37)			MCWBR	10.2	9.5	9.5	8.4	8.1	6.1	5.5	5.5	6.5	7.5	8.3	9.4	(37)
(38)		5% WB	MCDWB	16.1	15.8	16.9	17.2	16.4	13.8	14.0	13.4	14.0	14.5	14.8	15.0	(38)
(39)			MCWBR	10.1	8.7	8.5	7.6	6.9	5.3	5.4	5.5	6.5	7.4	8.0	9.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.391	0.439	0.467	0.513	0.503	0.439	0.423	0.427	0.442	0.427	0.396	0.389	(40)	
(41)		taud	2.165	2.007	1.932	1.841	1.896	2.118	2.173	2.169	2.127	2.164	2.218	2.192	(41)	
(42)		Ebn at Noon	258	256	258	251	254	270	274	272	263	257	255	252	(42)	
(43)		Edh at Noon	39	50	58	66	63	50	47	47	47	43	37	36	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1010	1281	1702	2066	2308	2485	2439	2227	1917	1490	1108	917	(44)	
(45)		RadStd	43	73	62	60	69	40	27	31	35	51	45	40	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	+0.76	N/A	N/A	N/A	+0.45	N/A	N/A	-112	+308	+187

Nomenclature: See separate page